

NZ Residential EV Charger — Detailed Electrician Guide

Mode 3 EVSE up to 32 A / ~7.4 kW single-phase and up to ~22 kW three-phase

Quick reference, location rules, compliance checklist, and worked examples

Document version: 1.0 | May 2026 | For registered electrical workers in New Zealand

Disclaimer: Educational field guide only — not legal advice. Mandatory requirements come from the Electricity (Safety) Regulations 2010 (ESR), cited AS/NZS 3000:2018 + Amendments 1–3 (legally cited from 13 November 2025 per SL 2025/225), and WorkSafe EV charging guidance. Purchase current standards; confirm distributor rules and CoC obligations before work.

Installer quick reference (tear-out summary)

Item	Requirement	Clause / source
Who may install	Registered electrical worker only; issue CoC	ESR; EWRB
Wiring Rules edition	AS/NZS 3000:2018 + Amd 1, 2, 3 (use now; mandatory for new work after 12 Nov 2026)	ESR Sch 2; WorkSafe Dec 2025
Domestic Mode 3/4	Dedicated final subcircuit; Type B 30 mA (all poles) OR Type A + IEC 62955 RDC-DD; direct fixed connection	7.9.3.3 (NZ)
Single-phase subcircuit	Minimum 32 A current-carrying capacity	7.9.3.3(a)
Three-phase subcircuit	7.9.3.3(a) states 32 A minimum for single-phase only — size 3-phase per EVSE rating + AS/NZS 3008; still apply 7.9.3.3(b)–(f)	7.9.3.3; 3.4
Height	Socket / EVSE cable outlet ≥ 800 mm above floor or ground	7.9.3.3(f)
Isolator	≥ 32 A, adjacent to charging facility	7.9.3.3(e); 2.3.2.2.1
Origin	From MEN switchboard (WorkSafe Jan 2026 addendum)	WorkSafe addendum §3
Mode 1	No socket-outlets for Mode 1 charging	7.9.2.2
Max demand	Assess full load; LMS often required for 32 A continuous	2.2.2; WorkSafe addendum
RCD test	Include DC leakage in function test where applicable	ESR mod. 8.3.10
Cable volt drop	≤ 5% PoS → EVSE (7% dedicated substation only)	AS/NZS 3000 CI 3.6.2
Supply at PoS	±10% of 230 V allowed on network — not cable VD	ESR Reg 28 (Nov 2025)
Product	SDoC; IEC 61851; 230/400 V 50 Hz; RDC-DD if Type A path	ESR; WorkSafe

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1. Regulatory stack

Instrument	Role for residential EVSE
Electricity Act 1992	Overarching electrical safety duties
Electricity (Safety) Regulations 2010	Installation work, product conformity, CoC
ESR Reg 59 + Schedule 2	Low-voltage installations must comply with Part 2 of cited AS/NZS 3000
AS/NZS 3000:2018 + Amd 1, 2, 3	Wiring Rules — Clause 7.9 (NZ only) for EV supplies
WorkSafe EV Charging Safety Guidelines (3rd ed.) + Jan 2026 Addendum	Regulator expectations (MEN board, load management, 32 A sockets)
AS/NZS 3008.1.1	Cable current capacity and voltage drop
IEC 61851-1 / -22	Mode 3 EVSE product safety
IEC 62955	RDC-DD (Type A alternative path)
SNZ PAS 6011:2023	Voluntary consumer/installer good practice (not cited in ESR)
Distributor Service & Installation Rules	Notification, capacity, controlled loads

Fact check: ESR citation update and transition dates verified against WorkSafe (19 Dec 2025) and EWRB guidance (Mar 2026).

2. Power levels: single-phase vs three-phase

Residential Mode 3 chargers in NZ are commonly supplied at 230 V single-phase or 400 V three-phase (line-to-line). The on-board charger limits what the vehicle accepts.

Supply	Typical installer setting	Approx. AC power	Notes
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1-phase, 16 A	16 A circuit (Mode 2 path 7.9.3.2 uses 20 A min.)	~3.7 kW	Lower demand; not full 7.9.3.3 single-phase 32 A path
1-phase, 32 A	32 A dedicated subcircuit	~7.4 kW	7.9.3.3(a) minimum capacity for single-phase Mode 3/4
3-phase, 16 A / phase	Subcircuit sized per load (e.g. 20 A or 32 A per phase per design)	~11 kW	Common “fast AC” residential cap; verify EVSE SDoC
3-phase, 32 A / phase	Heavy demand; submain and mains upgrade often required	~22 kW	Maximum typical residential AC; full max-demand study essential

Important distinction: Clause 7.9.3.3(a) explicitly requires a minimum **32 A current-carrying capacity for single-phase** final subcircuits. It does not state the same numeric minimum for three-phase. Three-phase installations must still comply with 7.9.3.3(b)–(f), general cable sizing (Section 3 / AS/NZS 3008), maximum demand (2.2.2), and WorkSafe principles (equipment rated 230 V or 400 V as applicable).

WorkSafe’s Jan 2026 addendum notes that with load management, a 32 A (7.5 kW) charger can often satisfy daily driving without exceeding typical domestic peak demand — but this is **not** permission to skip design.

3. Mandatory NZ rules (Clause 7.9)

3.1 Clause 7.9.3.3 — Domestic Mode 3 and Mode 4 (verified text)

Ref	Requirement
(a)	Single-phase final subcircuits: minimum 32 A current-carrying capacity
(b)	Final subcircuit shall supply no other point in the wiring
(c)	Separate Type B RCD 30 mA, all live conductors (active and neutral); RCBO allowed
(c) exc.	Type A 30 mA + RDC-DD per IEC 62955 (≥ 6 mA DC detected)
(d)	Direct connection (fixed wiring) — not a general-purpose socket circuit
(e)	Isolator $\geq$ 32 A , adjacent to charging facility (Clause 2.3.2.2.1)
(f)	Socket-outlet or EVSE cable $\geq$ 800 mm above floor/ground
NOTE	If equipment > 20 A single-phase: consider mains upgrade and/or load management

Source: AS/NZS 3000:2018 Amendment 1, Clause 7.9 (NZ only) — electrotraining / Standards extract.

3.2 Other Clause 7.9 items

- **7.9.2.2:** Socket-outlets shall **not** be installed for Mode 1 charging.
- **7.9.3.1:** New domestic installations with defined integral garage **shall** provide a Mode 2 or Mode 3/4 charging facility.
- **7.9.3.2:** Mode 2 — 20 A minimum dedicated subcircuit; approved socket types; 800 mm height.
- **7.9.4:** Non-residential — Type B (or Type A + RDC-DD for Mode 3) on final subcircuit.

3.3 Transition (ESR 2025)

When	Rule
From 13 Nov 2025	May use AS/NZS 3000:2018 + Amd 1–3 for alterations/new work
New construction after 12 Nov 2026	Must use cited 2018 edition + amendments
Alteration started after 13 Nov 2025, unfinished by 12 Nov 2026	CoC completed portion; balance to new standard

4. Protection: Type B vs Type A + RDC-DD

Board	EVSE	Compliance path
2-pole Type B RCBO/RCD 30 mA	Any compliant Mode 3	7.9.3.3(c) — simplest
2-pole Type A RCBO/RCD 30 mA	Built-in IEC 62955 RDC-DD (verify SDoC)	7.9.3.3 exception
EVSE with integrated Type B	Per manufacturer	WorkSafe addendum: no duplicate Type B required at board, but subcircuit still needs RCD coverage per 3000 note

Protection must cover the **entire final subcircuit** from origin to EVSE (cable between board and charger included). Single-pole devices are not acceptable where all poles must be protected.

RCD function testing per modified **8.3.10** must include DC leakage checks where applicable (2025 ESR). This verifies installation — not factory certification of the RCD-DD.

5. Where NOT to install (restricted zones)

Use this table on site before mounting. “Mandatory” items derive from AS/NZS 3000 or ESR; “Do not” items are strong guidance from Appendix P, WorkSafe, or related clauses.

Zone / condition	EVSE / socket?	Basis
Bathroom, shower, or bath Zone 0, 1, or 2 (wet areas per Section 6, Table 6.1)	Not permitted (socket-outlets excluded; EVSE is fixed electrical equipment — treat as prohibited in wet zones unless full IP and clause-by-clause compliance)	AS/NZS 3000 Section 6
Within 0.6 m of bath edge, 0.4 m above basin, shower Zone 2 without IPX4	Not permitted for standard accessories	Figs 6.1, 6.8, 6.13
Swimming pool zones 0, 1, 2	Not permitted without pool-specific compliance	Section 6 pool rules
Mode 1 charging from general socket-outlets	Prohibited — no sockets for Mode 1	7.9.2.2 (NZ)
Below 800 mm above floor or ground	Non-compliant for 7.9.3.2 / 7.9.3.3	7.9.3.3(f)
Shared final subcircuit with other loads (lights, power, HWS)	Non-compliant for Mode 3/4 domestic	7.9.3.3(b)
Hazardous area (flammable gas/vapour/dust) without appropriate Ex equipment	Not permitted unless classified and designed to AS/NZS 60079 series	Section 7 hazardous installations
Immediately behind vehicle bumper path, tight driveway turn, or unprotected car wash bay	Do not install without mechanical protection (bollards, barriers)	Appendix P (informative)
Outdoor exposed weather without adequate IP (see Section 6)	Non-compliant if IP rating inadequate for location	4.1.3; Section 6
Inside cupboard storing chemicals, pool chlorine, or flammable liquids (poor ventilation + corrosion)	Do not install — manufacturer environment limits	Risk assessment; PAS 6011 good practice
Where cable could be damaged by doors, gates, or vehicle doors	Do not install without mechanical protection	Appendix H / P
Detached garage subboard without confirmed MEN / outbuilding supply compliance	Do not energise until design verified	7.2.4.5; WorkSafe addendum
32 A continuous load on unknown/old consumer mains	Do not install without max-demand and volt-drop study	EWRB; WorkSafe addendum

5.1 Mode 2 vs Mode 3 location note

Mode 2 (IC-CPD plugged into socket) is discouraged by WorkSafe for employer vehicles and high sustained current. Mode 3 fixed EVSE is the preferred residential solution for 32 A equivalent charging.

6. Heights, clearances, and mounting

Dimension	Requirement	Type
Minimum height to socket / EVSE cable outlet	≥ 800 mm above floor or ground	Mandatory (7.9.3.3(f), 7.9.3.2(c))
Recommended ergonomic height	900–1 200 mm AFFL to centre of unit (cable hang, accessibility)	Good practice / manufacturer
Maximum practical height	Typically ≤ 1 500 mm unless ladder access agreed (maintenance, cable weight)	Good practice
Side clearance from wall corner	Often ≥ 300 mm from corner (manufacturer); allows door swing and cable bend	Manufacturer install manual
Vehicle reach	Charging port reachable without trailing cable across traffic path	Site layout
Outdoor IP	IP44 minimum typical (Appendix P informative); IP55+ recommended exposed rain	Appendix P; Section 6
Impact	IK07+ where vehicle impact foreseeable (car parks, tight garages)	Appendix P (informative)
Isolator	Adjacent to charging facility; visible or identifiable from EVSE	7.9.3.3(e)

7. Recommended locations and examples

7.1 Suitable location checklist

- ☐ Within dedicated garage or carport bay — wall facing vehicle charge port
- ☐ Exterior wall near parking with weather-rated EVSE (IP55+ if exposed)
- ☐ Close to MEN main switchboard when practical (short subcircuit, clear max-demand path)
- ☐ Protected from vehicle contact (bollard, wheel stop, offset from door travel)
- ☐ Ventilation and manufacturer temperature range met
- ☐ Distributor notification completed if required

7.2 Worked location examples

Example A — Integral garage (new build): Mode 3 wallbox on rear wall of garage, 1 000 mm AFFL, 400 mm from internal corner, IP44 minimum (dust/moisture). Dedicated 32 A subcircuit from garage MSB (MEN). Type A RCBO + EVSE RDC-DD. Vehicle parks nose-in; Type 2 tethered cable reaches port without crossing pedestrian door path. *Complies: 7.9.3.3, 7.9.3.1 facility provided.*

Example B — Carport (no full enclosure): IP55 wallbox on house wall facing carport slab, 900 mm AFFL, conduit in wall from MSB 6 m away. Concrete-mounted bollard in front of charger. Load management set to 16 A evening peak. *Complies if IP, height, protection, and max demand documented.*

Example C — Driveway beside house: Three-phase 11 kW EVSE near meter/MSB; 3-pole Type B RCBO; subcircuit sized 20 A/phase (or per EVSE). Height 850 mm; cable route in UV-stable conduit above ground level. 7.9.3.3(b)–(f)

apply; 32 A single-phase rule (a) N/A to 3-phase.

Example D — What to refuse: Customer requests charger inside ensuite bathroom Zone 2, 600 mm from basin.
Decline — wet zone prohibition; propose external garage or exterior IP-rated location outside classified zones.

Example E — Detached garage: 25 m submain; EVSE on front wall. Confirm MEN/outbuilding rules before choosing garage DB as subcircuit origin. RCBO at origin of final subcircuit; 32 A isolator at EVSE; submain sized for 32 A + other garage loads. *See Section 10 Example 5.*

8. Voltage drop and supply voltage — do not confuse 5% with 10%

Electricians often ask whether “voltage drop is now 10%.” The answer depends on **which rule** you mean:

Topic	Limit	Source	Applies to EV charger sizing?
Installation voltage drop (cable losses PoS → EVSE)	5% max (11.5 V at 230 V; 20 V line-to-line at 400 V). 7% only if supply is from a dedicated on-premises LV substation for that installation.	AS/NZS 3000:2018 Clause 3.6.2 (unchanged in Amd 1–3). ESR Reg 61 also references 5% for Part 1 certified designs.	Yes — size submains and 32 A EV final subcircuits to this limit using AS/NZS 3008.
Supply voltage tolerance at point of supply	±10% of nominal (was ±6%). Momentary fluctuations excepted.	ESR Regulation 28 , amended 13 November 2025 (SL 2025/225).	Indirectly — worst-case utilisation voltage may be higher/lower; does not allow 10% drop in your cables.
Extra-low voltage installations	10% voltage drop in conductors (ELV only).	AS/NZS 3000 Clause 7.5.7	No — residential 230/400 V Mode 3 EVSE is not ELV.

Design tip — 32 A EV, 25 m submain: Calculate volt drop on the full path (consumer mains + submain + final subcircuit) and keep the installation total within **5%** at maximum load. Separately, check the charger and vehicle will operate acceptably if supply at the meter is at the upper or lower end of the new **±10%** network band (roughly 207–253 V on a 230 V nominal single-phase supply).

Fact check May 2026: No published change to AS/NZS 3000 Clause 3.6.2 raising general LV installation voltage drop to 10%. If using AS/NZS 3000:2018 Rul 1:2024, confirm its scope against the purchased ruling — it does not replace Clause 3.6.2 unless explicitly stated there.

9. Cable, submain, and capacity

- Size cable per **AS/NZS 3008.1.1**: installation method, grouping, ambient, length.
- Installation voltage drop: ≤ **5%** from point of supply to EVSE (see Section 8).
- Single-phase 32 A: many installs use **6 mm²** Cu for short runs — always calculate; do not copy blindly.
- Three-phase: size each phase conductor for design current; neutral and earthing per 3000.
- **Maximum demand** (2.2.2): EV often at full rated current unless proven assessment, limitation, or LMS.
- Notify electricity distributor when required by network rules.

10. Testing, CoC, and records

1. Visual: location, height ≥ 800 mm, IP rating, labelling, dedicated circuit, isolator position.
2. Section 8: continuity, insulation, polarity, earth fault-loop impedance where required.
3. RCD operation including **DC leakage** per ESR-modified 8.3.10.
4. EVSE operational checks per manufacturer (earth monitoring, etc.).
5. Record energization date at switchboard (2018 edition).
6. Issue **Certificate of Compliance**; attach EVSE SDoC to CoC (EWRB good practice).

11. Worked installation examples

1. Single-phase 7.4 kW (32 A) — MSB in garage

63 A supply; 8 m run; Type 2 wallbox with RDC-DD; Type A 2-pole RCBO 32 A; 6 mm² V75; LMS throttles to 20 A when oven on. CoC references 7.9.3.3 and LMS settings.

2. Single-phase — old 50 A board, no spare capacity

Install 3.7 kW (16 A) on dedicated 20 A Mode 2 facility OR Mode 3 with LMS cap 12 A overnight OR supply upgrade. Do not fit 32 A hardware without assessment.

3. Three-phase 11 kW (16 A/phase)

400 V supply; 3-pole Type B 30 mA; cable 4 mm² per phase (verify); EVSE SDoC 400 V 50 Hz; max demand includes other 3-phase loads. 7.9.3.3(a) not applicable; (b)–(f) are.

4. Three-phase 22 kW (32 A/phase) — high-end home

Distributor upgrade likely; 10 mm² or larger per phase per 3008; main switch and submain upgraded; fixed load management mandatory; CoC documents full calculation.

5. Detached garage subboard

Verify MEN switchboard origin per WorkSafe; size 25 m submain; protection at final subcircuit origin; local isolator at EVSE; 8.3.9 outbuilding tests.

12. Master installer checklist

- ☐ Design to AS/NZS 3000:2018 + Amendments 1, 2, 3
- ☐ Site survey: restricted zones (Section 6 wet areas, impact, hazardous)
- ☐ Height ≥ 800 mm; IP/IK suitable for environment
- ☐ Dedicated final subcircuit; direct connection Mode 3
- ☐ Type B 2-pole 30 mA OR Type A + IEC 62955 RDC-DD (SDoC on file)
- ☐ Subcircuit from MEN switchboard (confirm detached garage case)
- ☐ Single-phase: 32 A min. subcircuit capacity; three-phase: sized per load + 7.9.3.3(b)–(f)
- ☐ 32 A isolator adjacent to EVSE
- ☐ Installation volt drop ≤ 5% (PoS → EVSE); supply ±10% is network only (Reg 28)
- ☐ Max demand / cable size calculated (AS/NZS 3008)
- ☐ Load management or upgrade if sustained 32 A exceeds capacity
- ☐ SDoC, IEC 61851, 230/400 V 50 Hz confirmed
- ☐ Section 8 tests + DC leakage; CoC issued
- ☐ Distributor notified if required

13. References (fact-checked May 2026)

1. AS/NZS 3000:2018 Amendment 1 — Clause 7.9 (NZ) — [electrotraining PDF](#)
2. Electricity (Safety) Regulations 2010 — [legislation.govt.nz](#)
3. ESR Reg 28 — voltage supply to installations ($\pm 10\%$) — [legislation.govt.nz](#)
4. Electricity (Safety) Amendment Regulations 2025 (SL 2025/225) — [legislation.govt.nz](#)
5. WorkSafe — ESR changes Dec 2025 — [worksafe.govt.nz](#)
6. WorkSafe — EV guidelines addendum Jan 2026 — [worksafe.govt.nz](#)
7. WorkSafe — EV charging safety guidelines (3rd ed.) — [worksafe.govt.nz](#)
8. EWRB — EV charging installations — [ewrb.govt.nz](#)
9. EWRB — Installation of EV charging equipment — [ewrb.govt.nz](#)
10. SNZ PAS 6011:2023 — Residential EV charging (voluntary) — [standards.govt.nz](#)
11. MBIE Build — Dedicated home charging units — [buildmagazine.org.nz PDF](#)

Companion summary: *NZ-Residential-EV-Charger-Installation-Guide.pdf* (shorter 32 A single-phase focus). Standards change — verify before use.